

C4.2 Identifying the products of chemical reactions

Summary

Types of substances can be classified according to their general physical and chemical properties. This section explores the tests that can be used to identify the products of reactions by looking at their physical and chemical properties.

Underlying knowledge and understanding

Learners should be familiar with cations and anions from their work on electrolysis.

Common misconceptions

Learners confuse mass and density so in reactions involving change of state, learners reason that the products from a precipitation reaction are heavier than the starting materials and that when a gas is produced the reaction has lost mass overall.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM4.2i ☒	interpret charts, particularly in spectroscopy	M4a

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
C4.2a (Combined Science C3.1g)	describe tests to identify selected gases	oxygen, hydrogen, carbon dioxide and chlorine		
C4.2b ☒	describe tests to identify aqueous cations and aqueous anions	calcium, copper, iron(II), iron(III) and zinc using sodium hydroxide; tests and expected results for carbonate ions (using diluted acid), chloride, bromide and iodide ions (using acidified dilute silver nitrate) and sulfate ions (using acidified dilute barium chloride or acidified barium nitrate)	WS1.4a	Tests for cations using sodium hydroxide. (PAG C5) Tests for anions using silver nitrate and barium sulfate. (PAG C5)
C4.2c ☒	describe how to perform a flame test		WS1.2b, WS1.2c, WS2a, WS2b	Flame tests. (PAG C5)
C4.2d ☒	identify species from test results			Testing unknown solutions for cations and anions. (PAG C5)
C4.2e ☒	interpret flame tests to identify metal ions	the ions of lithium, sodium, potassium, calcium and copper	WS1.4a	
C4.2f ☒	describe the advantages of instrumental methods of analysis	sensitivity, accuracy and speed	WS1.1e, WS1.2c, WS1.2d, WS1.2e	
C4.2g ☒	interpret an instrumental result given appropriate data in chart or tabular form, when accompanied by a reference set of data in the same form	the features of a mass spectroscopy chart	M4a	WS1.3e